

**INITIAL STATIC VERTICAL PILE LOAD TEST ON 900mm DIA PILE
FOR BDD CHAWLS REDEVELOPMENT PROJECT WORLI, MUMBAI.**

BUILDING NUMBER-3 (WING-B).

(INITIAL TEST PILE TP- 01)



Submitted to

CLIENT:-MHADA.

CONSULTANT:- GEOCON.

PMC :- VIVEK Bhole ARCHITECT PVT. LTD.

CONTRACTOR:- TATA PROJECTS.



ZedGeo Systems Pvt Ltd. Mumbai.

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**INITIAL STATIC VERTICAL PILE LOAD TEST ON 900mm DIA PILE
FOR BDD CHAWLS REDEVELOPMENT PROJECT WORLI, MUMBAI.
BUILDING NUMBER-3 (WING-B).
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INITIAL STATIC VERTICAL PILE LOAD TEST ON 900mm DIA PILE
FOR BDD CHAWLS REDEVELOPMENT PROJECT WORLI, MUMBAI.
BUILDING NUMBER-3 (WING-B).
(INITIAL TEST PILE TP- 01)

1.0 GENERAL

1.1 Clients decided to carry out static pile testing work on 900mm diameter pile to estimate load carrying capacity in vertical direction and settlement. M/s ZedGeo Systems Pvt Ltd., Mumbai was entrusted with work of static pile load test. This is the Initial test at BUILDING NUMBER-3 (WING-B) WORLI BDD CHAWLS, MUMBAI.

1.2 This report covers data for one vertical pile load test. This report covers calculation of safe load capacity for pile based on data collected during fieldwork.

1.3 The following codes of practices have been adopted.

- IS: 14593 – 1998 (Reaffirmed 2003). "Design and Construction of Bored Cast-in-Situ Piles Founded on Rocks – Guidelines."
- IS 2911 (Part 4) (Reaffirmed 2013) "Code of Practice for Design and Construction of Pile Foundations -Concrete Piles -Bored Cast – In – situ piles - Load Tests on Piles".

2.0 SCOPE OF WORK

Pile details are tabulated as below.

2.1 Pile details for Initial Pile (For Vertical Load Test)

Location: - Test Pile-1.

The details of the pile are as given below:

Maximum vertical Safe capacity of Pile = 420T

Diameter of Pile = 900mm

Grade of Concrete = M40

Pile Depth = 11.51m

Test Load

2.2 Vertical test load for Initial pile

The design vertical load on the pile is 420T.

The pile is required to be tested to a load of 1050T.

3.0 METHODOLOGY

3.1 The load testing on piles shall be conducted as per IS: 2911 (Part 4) – 1985 (Reaffirmed 2013).

Vertical Load Test On Piles

3.2 Test Load

The Initial load test was carried out to a test load of 2.5 times the design load as per clients requirement. The maximum test load was 1050T for test pile.

3.3 Initial Vertical Load Test on Piles

The pile load test was conducted by applying a series of vertical load on the test pile. The test pile has been loaded in increment up to the test loads and then unloaded as given in Table 3-1. The load was applied by means of 1 hydraulic jack of 1500 tons reacting against a symmetrically erected reaction crown system . The hydraulic jack was of adequate capacity and had a pressure gauge and remote control pump.

The pile to be tested was chipped and dressed to a well-levelled surface. All the sensors were fixed at the time of casting of pile and immediately after the dressing of pile. The circular steel plates of suitable thickness and of slightly bigger diameter than the diameter of the pile were placed over a fine layer of sand spread over the top of the pile.

Four jacks were placed diametrically opposite and places between the gap formed by the top of the plate resting on the pile and the lower plate of the crown. The jacks were connected and operated by one pump. The testing agency submitted calibration charts showing the correctness of the calibration of the pressure gauges and the dial gauges before use. All jacks will be fitted with locking devices. Another plate of suitable thickness shall be placed over the ram of the jack, which is later raised by operating the hydraulic pump so that the plate on the top of the ram butts against the bottom plate of the crown.

Reading of settlement and rebound was recorded with the help of four dial gauges of 0.01 mm sensitivity and resting on diametrically opposite ends of the pile cap. The dial gauges were fixed to a support at least 3 times the diameter of the pile or a minimum of 1.50 m away clear from the edge of pile. Readings on the dial gauges are to be observed immediately before and after application of loads, and immediately before and after release of loads.

IVPLT AT BUILDING NUMBER-3 (WING-B), 900MM (420T).

The loading was applied in increments of 20 % of the design load on Pile. Each load was kept for 1 Hr. During the unloading stages, the load on the pile shall be maintained for a minimum of 10 minutes and the subsequent elastic rebound in the pile shall be measured accurately by dial gauges. The final load was maintained for 24 hrs and the corresponding settlement was observed. Rebounding was recorded after the entire load is released. The pile test data is suitably presented by curves drawn between variables namely load and displacement and safe loads shown on the graphs including field observations.

The sequence of loading and unloading shall be as described below

1 jack ram area = 2551 cm².

so effective ram area is 2551 cm².

Design Load= 420T ,Test Load = 1050T.

Load Increment shall be 20% of Design load (420T), so 84T

As the least count of the pressure gauge is 10 kg/cm², exact 84MT.

of increment cannot be attained. Hence the values close to the incremental load is considered.

Table 1 – Load Sequence

Sr. no.	Pressure (kg/cm ²)	Load (MT)	Reading time (mins)
1.	0	0	0
2.	40	102.04	1,15,30,45,60 Mins
3.	80	204.08	1,15,30,45,60 Mins
4.	120	306.12	1,15,30,45,60 Mins
5.	160	408.16	1,15,30,45,60 Mins
6.	200	510.20	1,15,30,45,60 Mins
7.	240	612.24	1,15,30,45,60 Mins
8.	280	714.28	1,15,30,45,60 Mins
9.	320	816.32	1,15,30,45,60 Mins
10.	360	918.36	1,15,30,45,60 Mins
11.	400	1020.40	1,15,30,45,60 Mins
12	420	1071.42	1440 Mins
UNLOADING			
13	400	1020.40	1,5,15 Mins
14	360	918.36	1,5,15 Mins
15	320	816.32	1,5,15 Mins
16	280	714.24	1,5,15 Mins
17	240	612.24	1,5,15 Mins
18	200	510.20	1,5,15 mins
19	160	408.16	1,5,15 mins
20	120	306.12	1,5,15 mins
21	80	204.08	1,5,15 mins
22	40	102.04	1,5,15 mins
23	00	0	1,5,15 mins

4.0 RESULTS

4.1 Acceptance Criteria for Vertical Pile Load Test

The Safe Capacity of Piles is considered to be the least of the following as per IS: 2911, (Part 4):2013

- Two thirds of load at which total settlement attains a value of 12mm or maximum of 2 percent of the pile diameter which in this case works out to be 18mm, whichever is less.
- 50 % of the load corresponding to a settlement of 10% of pile diameter i.e 90mm

The Maximum settlement as per our field record at 1070.42T after 24 hours = 7.22 mm.

Total Rebond = 3.16 mm

The net settlement = 4.06 mm

So, as per the Test data and the graph we can say that the test pile has shown more load carrying capacity than design load of 420T.

So, 420T can be adopted as the safe vertical load for working piles.

11th Dec 2025

ZedGeo Systems Pvt Ltd, Mumbai.



Prasad Zantye
M.E.(Geotech), B.E.(Const.)

READINGS AND GRAPH



ZedGeo Systems Pvt. Ltd. Mumbai.

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Website:- www.zedgeosystems.com

ZedGeo Systems Pvt Ltd., Mumbai.

RECORD OF FOOTING LOAD TEST NO:-TP- 01

PROJECT:-BDD CHAWLS REDEVELOPMENT PROJECT

WORLD. MUMBAI.

LOCATION - Building Number-3 (Wing-B).

CONSULTANT :- GEOCON.

CLIENTS:- MHADA..

PMC :- VIVEK BHOLE ARCHITECT PVT. LTD.

CONTRACTOR:- TATA PROJECT.

lc of dial gauge:- 0.01mm

Type of Test:- IVPLT

Design Load :- 420 MT

Test Load :- 1050MT

Mixed Design :- M40.

Pile Diameter : - 900mm

Page:-1

Ram Area :- 2251cm²

Date of Casting :- 11-09-2025

Pile Length :- 11.51 mtr

[illegible]

[illegible]

[illegible]

[illegible]

Pile Length :- 11.51 mtr

[illegible]

ZedGeo Systems Private Limited., Mumbai.

RECORD OF FOOTING LOAD TEST NO:-TP- 01

PROJECT:-BDD CHAWLS REDEVELOPMENT PROJECT

WORLI, MUMBAI.

LOCATION - Building Number-3 (Wing-B).

CONSULTANT :- GEOCON.

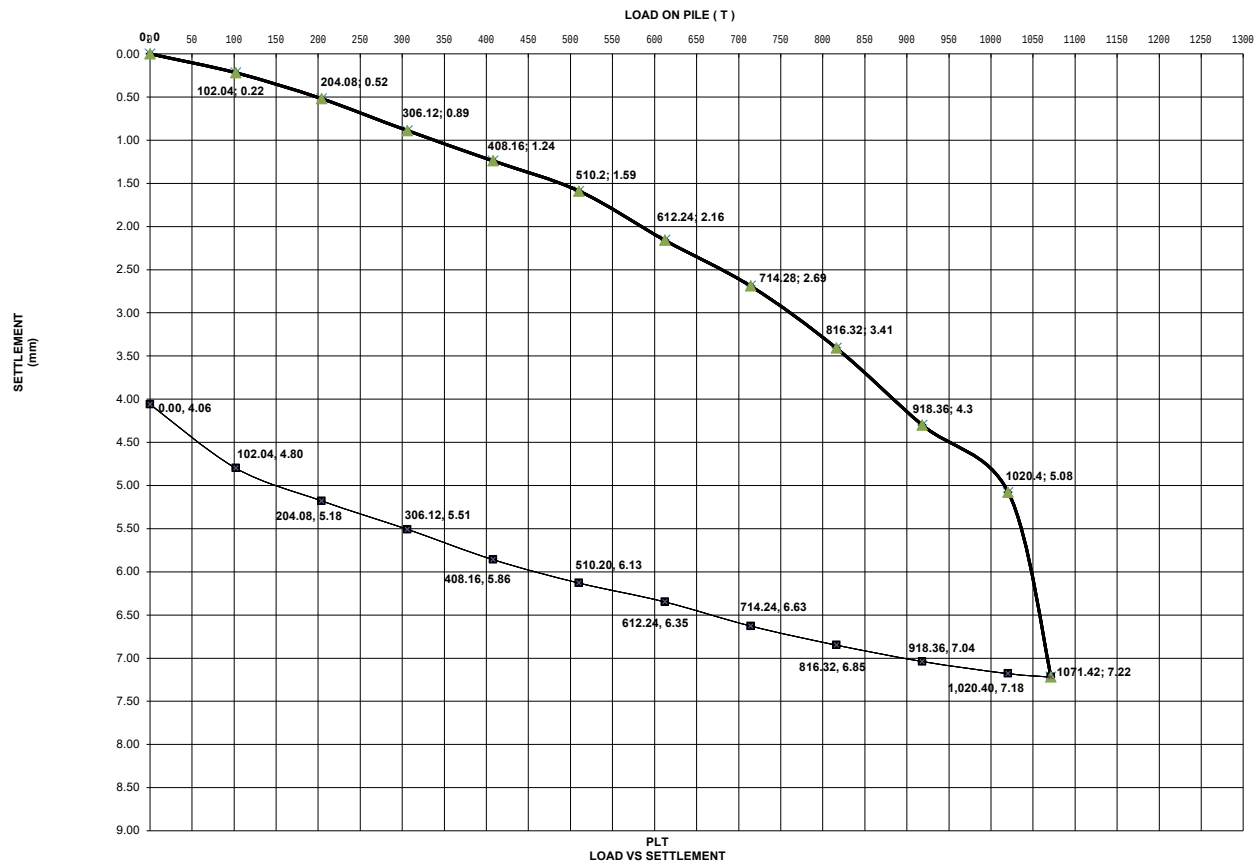
CLIENTS:- MHADA..

PMC :- VIVEK BHOLE ARCHITECT PVT. LTD.

CONTRACTOR:- TATA PROJECT.



LOADING			UNLOADING	
LOAD	AVERAGE DEFLECTION		LOAD	AVERAGE DEFLECTION
(T)	(mm)		(T)	(mm)
0.00	0.00		1071.42	7.22
102.04	0.22		1020.40	7.18
204.08	0.52		918.36	7.04
306.12	0.89		816.32	6.85
408.16	1.24		714.24	6.63
510.20	1.59		612.24	6.35
612.24	2.16		510.20	6.13
714.28	2.69		408.16	5.86
816.32	3.41		306.12	5.51
918.36	4.3		204.08	5.18
1020.4	5.08		102.04	4.80
1071.42	7.22		0	4.06



Maximum Settlement at 1071 T: 7.22 mm

Total Rebound : 3.16 mm

Net Settlement : 4.06 mm

ZedGeo Systems Pvt Ltd., Mumbai.
TATA/MHADA/GEOCON/ZED

TATA/MHADA/GEOCON/ZED
ZedGeo Systems Pvt Ltd., Mumbai.

FIELD READINGS



ZedGeo Systems Pvt. Ltd. Mumbai.

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**ZedGeo Systems Private Limited., Mumbai**

RECORD OF PILE LOAD TEST NO.: TP-01
PROJECT: BDD CHAWLS Redevelopment
Project, Worli Mumbai.
LOCATION: Building-3 Wing-
CLIENTS NAME: TATA Project
CONSULTANT:-
CONTRACTOR:-

L.C OF DIAL GAUGE:- 0.01mm
TYPE OF TEST:- IVPLT
DESIGN LOAD:- 420T
TEST LOAD:- 1050T
MIXED DESIGN:- M40
PILE DIAMETER:- 900mm

PAGE:- 1

RAM AREA:- 2551 cm²
DATE OF CASTING:- 11-9-25
PILE DEPTH:- 11.510Mtr.

DATE	TIME	PRESSURE	LOAD IN MT	Reading				Average settlement Test Pile	SIGNATURE			REMARK
				Reading 1	Reading 2	Reading 3	Reading 4		ZED	TATA	PMC	
9/12/25	(Hrs)	GAUGE READING kg/cm2										
	11:29	0	0	0	0	0	0	0	(RP)	Panathu.	H	*Loading*
	11:30	40	102.04	0.12	0.06	0.04	0.02	0.06	(RP)	Panathu.	H	
	11:45			0.19	0.13	0.10	0.10	0.13	(RP)	Panathu.	H	
	12:00			0.22	0.15	0.14	0.15	0.16	(RP)	Panathu.	H	
	12:15			0.24	0.16	0.17	0.18	0.18	(RP)	Panathu.	H	
	12:30			0.26	0.19	0.20	0.22	0.21	(RP)	Panathu.	H	
	12:31	80	204.08	0.45	0.40	0.39	0.38	0.40	(RP)	Panathu.	H	
	12:45			0.46	0.38	0.22	0.17	0.30	(RP)	Panathu.	H	
	13:00			0.47	0.41	0.43	0.41	0.43	(RP)	Panathu.	H	
	13:15			0.51	0.45	0.48	0.46	0.47	(RP)	Panathu.	H	
	13:30			0.55	0.49	0.53	0.51	0.52	(RP)	Panathu.	H	
	13:31	120	306.12	0.81	0.76	0.77	0.75	0.77	(RP)	Panathu.	H	
	13:45			0.84	0.80	0.80	0.79	0.80	(RP)	Panathu.	H	
	14:00			0.86	0.82	0.82	0.81	0.82	(RP)	Panathu.	H	
	14:15			0.89	0.83	0.86	0.85	0.85	(RP)	Panathu.	H	
	14:30			0.91	0.85	0.90	0.89	0.88	(RP)	Panathu.	H	
	14:31	160	408.16	1.32	1.20	1.27	1.21	1.25	(RP)	Panathu.	H	
	14:45			1.34	1.24	1.30	1.24	1.28	(RP)	Panathu.	H	
	15:00			1.35	1.26	1.35	1.27	1.30	(RP)	Panathu.	H	
	15:15			1.24	1.09	1.04	0.97	1.08	(RP)	Panathu.	H	
	15:30			1.34	1.20	1.25	1.16	1.23	(RP)	Panathu.	H	



ZedGeo Systems Private Limited., Mumbai

RECORD OF PILE LOAD TEST NO: TP-01
 PROJECT: BDD CHAWLS Redevelopment
 LOCATION: Projects work, Mumbai
Building No-3 (wing B)
 CLIENTS NAME: TATA Project
 CONSULTANT: TATA Project
 CONTRACTOR:

L.C OF DIAL GAUGE: 0.01mm
 TYPE OF TEST: I VPLT
 DESIGN LOAD: 480T
 TEST LOAD: 1050T
 MIXED DESIGN: M40
 PILE DIAMETER: 900mm

PAGE: 2

RAM AREA: 2551cm²
 DATE OF CASTING: 11-9-25
 PILE DEPTH: 11.50 Mtr

DATE	TIME	PRESSURE	LOAD IN MT	Reading				Average settlement	SIGNATURE			REMARK
	(Hrs)	GAUGE READING kg/cm2		Reading 1	Reading 2	Reading 3	Reading 4	Test Pile	Zed	TATA	PMC	
9/12/25	15:31	200	510.20	1.56	1.58	1.45	1.44	1.50	(R)	P. D. N. K.	P. D. N. K.	*Loading*
	15:45			1.57	1.60	1.43	1.42	1.50	(R)	P. D. N. K.	P. D. N. K.	
	16:00			1.59	1.62	1.48	1.43	1.53	(R)	P. D. N. K.	P. D. N. K.	
	16:15			1.61	1.65	1.53	1.46	1.56	(R)	P. D. N. K.	P. D. N. K.	
	16:30			1.65	1.67	1.55	1.49	1.59	(R)	P. D. N. K.	P. D. N. K.	
	16:31	240	612.24	2.21	2.12	2.04	1.96	2.08	(R)	P. D. N. K.	P. D. N. K.	
	16:45			2.26	2.15	2.07	1.97	2.11	(R)	P. D. N. K.	P. D. N. K.	
	17:00			2.26	2.17	2.09	1.98	2.12	(R)	P. D. N. K.	P. D. N. K.	
	17:15			2.28	2.19	2.09	1.98	2.13	(R)	P. D. N. K.	P. D. N. K.	
	17:30			2.28	2.24	2.11	2.01	2.16	(R)	P. D. N. K.	P. D. N. K.	
	17:31	280	714.28	2.72	2.75	2.57	2.58	2.65	(R)	P. D. N. K.	P. D. N. K.	
	17:45			2.72	2.75	2.58	2.58	2.657	P. D. N. K.	P. D. N. K.	P. D. N. K.	
	18:00			2.73	2.76	2.60	2.60	2.672	P. D. N. K.	P. D. N. K.	P. D. N. K.	
	18:15			2.75	2.77	2.61	2.60	2.68	P. D. N. K.	P. D. N. K.	P. D. N. K.	
	18:30			2.75	2.78	2.61	2.66	2.68	(R)	P. D. N. K.	P. D. N. K.	
	18:31	320	816.32	3.40	3.43	3.15	3.21	3.30	(R)	P. D. N. K.	P. D. N. K.	
	18:45			3.43	3.45	3.18	3.24	3.32	(R)	P. D. N. K.	P. D. N. K.	
	19:00			3.45	3.48	3.20	3.27	3.35	(R)	P. D. N. K.	P. D. N. K.	
	19:15			3.48	3.50	3.23	3.30	3.37	(R)	P. D. N. K.	P. D. N. K.	
	19:30			3.51	3.53	3.25	3.33	3.40	(R)	P. D. N. K.	P. D. N. K.	



**ZedGeo Systems Private Limited., Mumbai**

RECORD OF PILE LOAD TEST NO:- TP-01

PROJECT:- BOD CHAWL Redveliment

Project work Mumbai

LOCATION:- Building No3 (wing B)

CLIENTS NAME:- TATA project

CONSULTANT:-

CONTRACTOR:-

LC OF DIAL GAUGE:- 0.01

TYPE OF TEST:- IVPLT

DESIGN LOAD:- 420T

TEST LOAD:- 1050T

MIXED DESIGN:- m40

PILE DIAMETER:- 900mm

PAGE:- 4

HAM AREA:- 2551 cm²

DATE OF CASTING:- 11/9/2025

PILE DEPTH:- 10.50 mtr

DATE	TIME	PRESSURE	LOAD IN MT	Reading				Average settlement	SIGNATURE			REMARK
	(Hrs)	GAUGE READING kg/cm ²		Reading 1	Reading 2	Reading 3	Reading 4	Test Pile	ZED		PMC	
9/12/25												*Holding*
	22.30	420	1071.42	5.55	5.63	5.19	5.24	5.40	Patil	Panath	Panath	
	23.30			5.57	5.67	5.24	5.29	5.44	Patil	Panath	Panath	
	24.30			5.62	5.71	5.32	5.34	5.49	Patil	Panath	Panath	
10/12/25	1.30			5.65	5.74	5.35	5.37	5.52	Patil	Panath	Panath	
	2.30			5.69	5.79	5.41	5.41	5.57	Patil	Panath	Panath	
	3.30			5.72	5.81	5.43	5.43	5.60	Patil	Panath	Panath	
	4.30			5.75	5.84	5.46	5.44	5.63	Patil	Panath	Panath	
	5.30			5.80	5.87	5.49	5.47	5.65	Patil	Panath	Panath	
	6.30			5.81	5.88	5.50	5.48	5.66	Patil	Panath		
	7.30			5.81	5.89	5.52	5.50	5.68	Patil	Panath		
	8.30			5.85	5.92	5.60	5.61	5.74	Patil	Panath		
	9.30			5.89	5.95	5.66	5.66	5.79	Patil	Panath		

**ZedGeo Systems Private Limited., Mumbai**

RECORD OF PILE LOAD TEST NO: TP-01
PROJECT: BOD CHAWL - Redevelopment
Project work, mumbai
LOCATION: Building NO3 (wing B)
CLIENTS NAME:
CONSULTANT:- TATA Project
CONTRACTOR:-

L.C OF DIAL GAUGE:- 0.01
TYPE OF TEST:- IVPLT
DESIGN LOAD:- 420T
TEST LOAD:- 1050T
MIXED DESIGN:- m 40
PILE DIAMETER 2100mm

PAGE: 5

RAM AREA:- 2551 cm²
DATE OF CASTING:- 11/9/2025
PILE DEPTH:- 10.50mtr

DATE	TIME	PRESSURE	LOAD IN MT	Reading				Average settlement	SIGNATURE			REMARK
10/12/25	(Hrs)	GAUGE READING kg/cm ²		Reading 1	Reading 2	Reading 3	Reading 4	Test Pile	ZED	TPC	PMC	*Holding*
	10:30	420	1071.42	6.25	6.27	5.94	6.10	6.14	Pull	Faunak	✓	
	11:30			6.62	6.66	6.53	6.70	6.62	RR	Faunak	✓	
	12:30			6.80	6.88	6.85	6.90	6.85	RR	Faunak	✓	
	13:30			6.86	6.94	6.90	7.10	6.95	(R.P.)	Faunak	✓	
	14:30			6.94	6.95	6.95	7.15	6.99	RR	Faunak	Faunak	
	15:30			7.06	7.09	7.11	7.23	7.12	(R.P.)	Faunak	Faunak	
	16:30			7.10	7.14	7.17	7.26	7.16	(R.P.)	Faunak	Faunak	
	17:30			7.12	7.18	7.19	7.28	7.19	(R.P.)	Faunak	Faunak	
	18:30			7.13	7.20	7.20	7.31	7.21	(R.P.)	Faunak	Faunak	
	19:30			7.15	7.21	7.21	7.31	7.22	(R.P.)		Faunak	
	20:30			7.15	7.22	7.21	7.31	7.22	(R.P.)		Faunak	
	21:30			7.15	7.22	7.21	7.31	7.22	(R.P.)		Faunak	



ZedGeo Systems Private Limited., Mumbai

RECORD OF PILE LOAD TEST NO: TP-01
PROJECT:- BDD CHAWL Redevelopment
Project work Mumbai.
LOCATION:- Building No-3 (wing-B)
CLIENTS NAME:- TATA PROJECT
CONSULTANT:-
CONTRACTOR:-

LC OF DIAL GAUGE:- 0.01 mm
TYPE OF TEST:- I VPLT
DESIGN LOAD:- 420T
TEST LOAD:- 1050T
MIXED DESIGN:- M40
PILE DIAMETER:- 900 mm

PAGE: 6

RAM AREA:- 2551 cm²
DATE OF CASTING:- 11-9-25
PILE DEPTH:- 10.50 mtr.

DATE	TIME	PRESSURE	LOAD IN MT	Reading				Average settlement	SIGNATURE			REMARKS
	(Hrs)	GAUGE READING kg/cm ²		Reading 1	Reading 2	Reading 3	Reading 4	Test Pile	ZED	TATA	PMC	
10-12-25	21:31	400	1020.40	7.11	7.18	7.17	7.26	7.18	(R.P.)			*UNLOADING*
	21:35			7.11	7.18	7.17	7.26	7.18	(R.P.)			
	21:45			7.11	7.18	7.17	7.29	7.18	(R.P.)			
	21:46	360	918.36	7.00	7.08	7.06	7.20	7.08	(R.P.)			
	21:50			6.98	7.05	7.04	7.17	7.06	(R.P.)			
	22:00			6.97	7.03	7.02	7.14	7.04	(R.P.)			
	22:01	320	816.32	6.82	6.88	6.81	6.97	6.87	(R.P.)			
	22:05			6.81	6.84	6.80	6.94	6.84	(R.P.)			
	22:15			6.81	6.84	6.80	6.94	6.84	(R.P.)			
	22:16	280	714.24	6.63	6.67	6.59	6.76	6.66	(R.P.)			
	22:20			6.60	6.66	6.56	6.74	6.64	(R.P.)			
	22:30			6.59	6.66	6.55	6.73	6.63	(R.P.)			
	22:31	240	612.24	6.38	6.43	6.33	6.53	6.41	(R.P.)			
	22:35			6.33	6.38	6.27	6.50	6.37	(R.P.)			
	22:45			6.30	6.35	6.25	6.49	6.34	(R.P.)			
	22:46	200	510.20	6.11	6.17	6.01	6.28	6.14	(R.P.)			
	22:50			6.10	6.15	6.00	6.27	6.13	(R.P.)			
	23:00			6.10	6.15	6.00	6.27	6.13	(R.P.)			



**ZedGeo Systems Private Limited., Mumbai**

PAGE- 7

RECORD OF PILE LOAD TEST NO: TP-01
 PROJECT:- BDD CHAWL Redevelopment
 Project work, Mumbai.
 LOCATION:- Building No-3 (wing-B)
 CLIENTS NAME:- TATA PROJECT
 CONSULTANT:-
 CONTRACTOR:-

LC OF DIAL GAUGE:- 0.01 mm
 TYPE OF TEST:- IVPLT
 DESIGN LOAD:- 420T
 TEST LOAD:- 1050T
 MIXED DESIGN:- M40
 PILE DIAMETER:- 900mm

RAM AREA:- 2551 cm²
 DATE OF CASTING:- 11-9-25
 PILE DEPTH:- 10.50 Mtr.

DATE	TIME	PRESSURE	LOAD IN MT						Average settlement	SIGNATURE			REMARK
10-12-25	TIME (Hrs)	GAUGE READING kg/cm ²		Reading				Test Pile		ZED	TATA	PMC	*UNLOADING*
				Reading 1	Reading 2	Reading 3	Reading 4						
	23:01	160	408.16	5.85	5.90	5.77	6.03	5.88	(R.P.)			Ravi	
	23:05			5.82	5.88	5.76	6.00	5.86	(R.P.)			Ravi	
	23:15			5.81	5.88	5.75	5.99	5.85	(R.P.)			Ravi	
	23:16	120	306.12	5.53	5.61	5.46	5.71	5.57	(R.P.)			Ravi	
	23:20			5.49	5.57	5.41	5.67	5.53	(R.P.)			Ravi	
	23:30			5.48	5.55	5.38	5.64	5.50	(R.P.)			Ravi	
	23:31	80	204.08	5.20	5.29	5.15	5.36	5.25	(R.P.)			Ravi	
	23:35			5.17	5.24	5.12	5.31	5.21	(R.P.)			Ravi	
	23:45			5.15	5.21	5.09	5.28	5.18	(R.P.)			Ravi	
	23:46	40	102.04	4.89	4.92	4.80	4.97	4.89	(R.P.)			Ravi	
	23:50			4.83	4.87	4.75	4.81	4.84	(R.P.)			Ravi	
11-12-25	00:00			4.78	4.84	4.71	4.86	4.80	(R.P.)			Ravi	
	00:01	0	0	4.11	4.23	3.99	4.23	4.13	(R.P.)			Ravi	
	00:05			4.08	4.21	3.95	4.18	4.10	(R.P.)			Ravi	
	00:15			4.05	4.17	3.90	4.12	4.06	(R.P.)			Ravi	

Maximum Settlement at 1071.4 T $\Rightarrow$ 7.22 mm.Rebound $\Rightarrow$ 3.16 mm.Net Settlement $\Rightarrow$ 4.06 mm.

CALIBRATION CERTIFICATES



ZedGeo Systems Pvt. Ltd. Mumbai.

Tel:- 8097256577 Cell :- 9103622195/09821065636

E-mail:- zedgeotech@yahoo.com

Website:- www.zedgeosystems.com



Format No: PLTJ/01

Rev. No.- 00

CALIBRATION CERTIFICATE

CALIBRATION CERTIFICATE NO	: QCC-254-CR-02
CALIBRATION TEST REPORT DATE	: 26/02/2025
NAME OF CUSTOMER	: M/s. ZEDZEO SYSTEMS PRIVET LIMITED
PROJECT SITE ADDRESS	: NAVI MUMBAI
DATE OF CALIBRATION	: 26/02/2025
SUGGESTED NEXT CALIBRATION DUE DATE (As customer Requested)	25/02/2026

Details of Device Under Calibration (DUC)

Instrument Description : Hydraulic Pile Test Jack	Jack Model No. : 1500/350
Make : DYNAMIC	Range of Jack (Ton) : 1500
Tag / ID No : ZCB/HPTJ/02	Make pf pressure Gauge : Baumer
Power Pack Sr. No. : ZED01	Range of Pressure Gauge (Kg/cm ²) : 700
Pushing Jack Sr. No. : 22124717	Sr. No of Pressure Gauge : N164.59-00129
Ram Area of Jack (cm ²): 2551.17	Resolution of Pressure Gauge (Kg/cm ²) : 10

Technical Details of Hydraulic Jack

Outer Dia, mm	Close Length of Jack , mm	Dia of Piston , mm	Max Stroke , mm
600	650	425	350

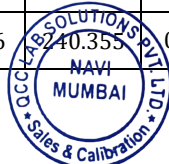
DETAIL OF MASTER EQUIPMENT USED FOR CALIBRATION

Id of Instrument	Range of Master	Calibrated By	Calibration Certificate No.	Calibration Date	Calibration Due Date
QCC/PGC/01	0- 700 bar	IDEMI , Mumabi	CC/PRL/ 0118 /25	27/01/2025	26/01/2026

CALIBRATION RESULTS

Calibration Method : DKD R-6 : 2016, SOP-CAL-10	Calibration done at Room Temp(° C) : 29.4
Calibration Done at Location : At Lab	Calibration done at Humidity(% RH) : 55
Pressure Transmitting Medium : Hydraulic	Certified Range of Hydraulic Jack (Ton) : 1479.34

Reading on DUC (kg/cm ²)	Reading on Standard (kg/cm ²)				Average Reading (Mean) (kg/cm ²)	Error (kg/cm ²)	Repeatability (kg/cm ²)	Actual Load in Ton (Given)	Solid Ram Area, cm ²	Pressure in Tons (Indicated)	Efficiency Of Jack
P (ACT)	M1	M2	M3	M4	[M1+M2+M3+M4]/4	Mean-P(Act)	Max Value ((M3-M1), (M4-M2))				
0	0.00	0.00	0.00	0.00	0.000	0.0000	0.000	0.00	2551.17	0.00	0.00
60	60.12	60.08	60.08	60.22	60.125	0.1250	0.140	153.07	2551.17	153.39	99.79
120	120.12	120.15	120.16	120.24	120.450	0.4500	0.090	306.14	2551.17	307.29	99.63
180	180.09	180.05	180.45	180.22	180.240	0.2400	0.360	459.21	2551.17	459.82	99.87
240	240.15	240.55	240.16	240.56	240.355	0.3550	0.010	612.28	2551.17	613.19	99.85





300	300.12	300.08	300.19	300.45	300.210	0.2100	0.370	765.35	2551.17	765.89	99.93
360	360.45	360.46	360.51	360.42	360.460	0.4600	0.060	918.42	2551.17	919.59	99.87
420	420.22	420.16	420.15	420.13	420.165	0.1650	-0.030	1071.49	2551.17	1071.91	99.96
480	480.24	480.26	480.24	480.45	480.298	0.2975	0.190	1224.56	2551.17	1225.32	99.94
540	540.22	540.08	540.65	540.25	540.300	0.3000	0.430	1377.63	2551.17	1378.40	99.94
580	580.45	580.61	580.69	580.64	580.598	0.5975	0.240	1479.68	2551.17	1481.20	99.90
PERMISSIBLE % OF EFFICIENCY >95 %									Average value:		99.87

Remarks:

1. The Reported Expanded uncertainty is stated as a standard uncertainty multiplied by a coverage confidence level factor $k=2$ at $\pm 95\%$

Note:

2. DUC stands for device under calibration.

3. The certificate shall refer only to the particular item submitted for calibration.

4. The certificate shall not be reproduced except in full unless written permission for the publication of an approved abstract has been obtained from the technical manager of QCC lab solution Pvt. Ltd. Navi Mumbai.

5. As found ; As left

6. The calibration results reported in the certificate are valid at the time of and under the stated conditions of measurement.

7. Calibration point done as per customer request



(Authorised Signatory)



QCC LAB

SOLUTIONS PVT. LTD.
Equipment Sales & Calibration Services

Format No:- QCC-PG-01

CALIBRATION CERTIFICATE

Rev. No:- 00

1. General Details									
Date of Receipt	:- 22/08/2025				Certificate Issue date	:- 22/08/2025			
Service Request No.	:- QCC-97				Certificate No.	:- QCC-97-CR-01			
Condition of Instruments	:- Satisfactory				Calibration to be done at	:- AT LAB			
Date of Calibration	:- 22/08/2025				Environmental Condition °C &	:- 27.5 °C & 53% rh			
Due date of calibration (As per customer requirement)	:- 21/08/2026				Reference Standard	:- DKD R-6 : 2016, QCC-SOP-4			
2. Customer Details									
Name of Customer	:- M/s. ZedGeo System Pvt. Ltd.								
Site Address	:- 4/8, Kamma CHS KG Marg, Prabhadevi, Mumbai-400025								
3. Details of Equipment									
Instrument Description	:- Pressure Gauge				Tag / ID No	:- SE/3942/01			
Scale Range(kg/cm ²)	:- 1000				Type of Pressure Gauge	:- Analog			
Resolution (kg/cm ²)	:- 10				MU Requirement(%)	:- NA			
Make	:- Baumer				Certified Range(kg/cm ²)	:- 700			
Model/ Sr. No.:	:- SS316L/ N164.59-00129				Pressure Transmitting Medium	:- Hydraulic			
4. Detail of Master Equipment									
Master Equipement Description	ID Mark		Calibration Certificate No.		Range of Master		Calibration Date		Calibration Due Date
Pressure Gauge	QCC/PG/01		CC/PRL/ 0305 /21-22		700 bar		19/10/2024		18/10/2025
5. Result									
Reading on DUC (kg/cm ²)	Reading on Standard (kg/cm ²)				Average Reading (Mean) (kg/cm ²)	Error (kg/cm ²)	Repeatability (kg/cm ²)	Hysteresis (kg/cm ²)	Expanded Uncertainty bar (±)
P (ACT)	M1	M2	M3	M4	[M1+M2+M3+M4]/4	Mean-P(Act)	Max Value ((M3-M1), (M4-M2))	Max Value ((M2-M1), (M4-M3))	
0	0.00	0.00	0.00	0.00	0.00	0.00	0.000	0.00	0.31
70	70.02	70.03	70.04	70.02	70.03	0.03	0.020	0.01	0.31
140	140.01	140.12	140.08	140.09	140.08	0.08	0.069	0.11	0.31
210	210.15	210.19	210.14	210.16	210.16	0.16	-0.010	0.04	0.31
280	280.22	280.19	280.16	280.19	280.19	0.19	0.000	0.03	0.31
350	350.26	350.29	350.24	350.26	350.26	0.26	-0.020	0.03	0.31
420	420.35	420.49	420.40	420.41	420.41	0.41	0.050	0.14	0.31
490	490.48	490.41	490.46	490.47	490.46	0.46	0.060	0.01	0.31
560	560.51	560.55	560.53	560.55	560.54	0.53	0.020	0.04	0.31
630	630.61	630.63	630.58	630.58	630.60	0.60	-0.030	0.02	0.31
700	700.76	700.78	700.79	700.70	700.76	0.76	0.030	0.02	0.31

(Calibration by)



(Authorized by)

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QCC LAB

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Sr. No	Determination	Value (Kgcm2)	% FSD
1	Maximum Error	1.12	0.16
2	Maximum Hysteresis	0.14	0.02

Remarks:

1. The Reported expanded uncertainty is evaluated at approximately 95% confidence level having Coverage factor k=2.

Note:

1. DUC stands for device under calibration.
2. The certificate shall refers only to the perticular item submitted for calibration .
3. The certificate shall not be reproduced except in full unless written permission for the publication of an approved abstract has been obtained from the technical manager of QCC lab solution Pvt. Ltd. Navi Mumbai.
4. As found ;As left
5. The calibration results reported in the certificate are valid at the time of and under the stated conditions of measurement.

(Calibration by)



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QCC LAB

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Equipment Sales & Calibration Services

Format No: QCC/DG/01

Rev. No. - 00

CALIBRATION CERTIFICATE

Calibration Certificate No.	: QCC-2303-16030
Calibration Report Date	: 01/05/2025
Customer Name	: M/s. ZEDZEO SYSTEMS PRIVET LIMITED
Site Address	: Navi Mumbai
Date of Calibration	: 01/05/2025
Calibration Due Date (as per customer requirment)	: 01/05/2026

DETAILS OF UNIT UNDER CALIBRATION

Equipement Description	: Analog Dial Gauge
Id of UUC	: ZSPL/DG/02
Make.	: BAKER
Model No.	: FJD964
Range (mm)	: 0-25 mm
Resolution (mm)	: 0.01

DETAIL OF MASTER EQUIPMENT USED FOR CALIBRATION

Master Equipement Description	Range	Calibration Certificate No.	Make	Calibration Date	Calibration Date
Dial calibration Tester	0-25 mm	M-210209-25-1	Reddy Instruments	05/09/2024	05/09/2025

Calibration Method	: IS 2092 -1983, QCC/SOP/15
Calibration Done on Location	: AT LAB
Room Temp. (°C) & Humadity (%RH)	: 20.1 & 56
Unit of Measurement : mm	

Sr.No.	Set point on DUC	Reading on master (Avg.)	Deviation/Error	Expanded Uncertainty in ±
1	0.0	0.0000	0.0000	0.007
2	2.5	2.5003	0.0003	
3	5.0	5.0009	0.0009	
4	7.5	7.5085	0.0085	
5	10.0	10.0013	0.0013	
6	12.5	12.5019	0.0019	
7	15.0	15.0064	0.0064	
8	17.5	17.5068	0.0068	
9	20.0	20.0084	0.0084	
10	22.5	22.5093	0.0093	
11	25.0	25.0092	0.0092	

Remarks:

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6. Calibration point don as per customer request

(Calibrated By)



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QCC LAB

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Format No: QCC/DG/01

Rev. No.- 00

CALIBRATION CERTIFICATE

Calibration Certificate No.	: QCC-2303-16031
Calibration Report Date	: 01/05/2025
Customer Name	: M/s. ZEDZEO SYSTEMS PRIVET LIMITED
Site Address	: Navi Mumbai
Date of Calibration	: 01/05/2025
Calibration Due Date (as per customer requirment)	: 01/05/2026

DETAILS OF UNIT UNDER CALIBRATION

Equipement Description	: Analog Dial Gauge
Id of UUC	: ZSPL/DG/01
Make.	: BAKER
Model No.	: JCB525
Range (mm)	: 0-25 mm
Resolution (mm)	: 0.01

DETAIL OF MASTER EQUIPMENT USED FOR CALIBRATION

Master Equipement Description	Range	Calibration Certificate No.	Make	Calibration Date	Calibration Date
Dial calibration Tester	0-25 mm	M-210209-25-1	Reddy Instruments	05/09/2024	05/09/2025

Calibration Method	: IS 2092 -1983, QCC/SOP/15
Calibration Done on Location	: AT LAB
Room Temp. (°C) & Humadity (%RH)	: 20.4 & 53
Unit of Measurement : mm	

Sr.No.	Set point on DUC	Reading on master (Avg.)	Deviation/Error	Expanded Uncertainty in ±
1	0.0	0.0000	0.0000	0.007
2	2.5	2.4984	-0.0016	
3	5.0	4.9992	-0.0008	
4	7.5	7.4968	-0.0032	
5	10.0	9.9983	-0.0017	
6	12.5	12.4846	-0.0154	
7	15.0	14.9854	-0.0146	
8	17.5	17.4837	-0.0163	
9	20.0	19.9914	-0.0086	
10	22.5	22.4911	-0.0089	
11	25.0	24.9930	-0.0070	

Remarks:

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Equipment Sales & Calibration Services

Format No: QCC/DG/01

Rev. No. - 00

CALIBRATION CERTIFICATE

Calibration Certificate No.	: QCC-2303-16032
Calibration Report Date	: 01/05/2025
Customer Name	: M/s. ZEDZEO SYSTEMS PRIVET LIMITED
Site Address	: Navi Mumbai
Date of Calibration	: 01/5/2025
Calibration Due Date (as per customer requirment)	: 01/05/2026

DETAILS OF UNIT UNDER CALIBRATION

Equipement Description	: Analog Dial Gauge
Id of UUC	: ZSPL/DG/02
Make.	: BAKER
Model No.	: HLB485
Range (mm)	: 0-25 mm
Resolution (mm)	: 0.01

DETAIL OF MASTER EQUIPMENT USED FOR CALIBRATION

Master Equipement Description	Range	Calibration Certificate No.	Make	Calibration Date	Calibration Date
Dial calibration Tester	0-25 mm	M-210209-25-1	Reddy Instruments	05/09/2024	05/09/2025

Calibration Method	: IS 2092 -1983, QCC/SOP/15
Calibration Done on Location	: AT LAB
Room Temp. (°C) & Humadity (%RH)	: 20.1 & 56
Unit of Measurement : mm	

Sr.No.	Set point on DUC	Reading on master (Avg.)	Deviation/Error	Expanded Uncertainty in ±
1	0.0	0.0000	0.0000	0.007
2	2.5	2.5003	0.0003	
3	5.0	5.0009	0.0009	
4	7.5	7.5085	0.0085	
5	10.0	10.0013	0.0013	
6	12.5	12.5019	0.0019	
7	15.0	15.0064	0.0064	
8	17.5	17.5068	0.0068	
9	20.0	20.0084	0.0084	
10	22.5	22.5093	0.0093	
11	25.0	25.0092	0.0092	

Remarks:

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Format No: QCC/DG/01

Rev. No.- 00

CALIBRATION CERTIFICATE

Calibration Certificate No.	: QCC-2303-16034
Calibration Report Date	: 01/05/2025
Customer Name	: M/s. ZEDZEO SYSTEMS PRIVET LIMITED
Site Address	: Navi Mumbai
Date of Calibration	: 01/05/2025
Calibration Due Date (as per customer requirment)	: 01/05/2026

DETAILS OF UNIT UNDER CALIBRATION

Equipement Description	: Analog Dial Gauge
Id of UUC	: ZSPL/DG/04
Make.	: BAKER
Model No.	: EJB670
Range (mm)	: 0-25 mm
Resolution (mm)	: 0.01

DETAIL OF MASTER EQUIPMENT USED FOR CALIBRATION

Master Equipement Description	Range	Calibration Certificate No.	Make	Calibration Date	Calibration Date
Dial calibration Tester	0-25 mm	M-210209-25-1	Reddy Instruments	05/09/2024	05/09/2025

Calibration Method	: IS 2092 -1983, QCC/SOP/15
Calibration Done on Location	: AT LAB
Room Temp. (°C) & Humadity (%RH)	: 20.3 & 56
Unit of Measurement : mm	

Sr.No.	Set point on DUC	Reading on master (Avg.)	Deviation/Error	Expanded Uncertainty in $\pm$
1	0.0	0.0000	0.0000	0.007
2	2.5	2.4991	-0.0011	
3	5.0	4.9995	-0.0005	
4	7.5	7.4976	-0.0024	
5	10.0	9.9997	-0.0003	
6	12.5	12.4967	-0.0033	
7	15.0	14.9941	-0.0059	
8	17.5	17.4936	-0.0064	
9	20.0	19.9985	-0.0015	
10	22.5	22.4969	-0.0031	
11	25.0	24.9990	-0.0010	

Remarks:

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